Sho Sakai

酒井　彰

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Summary

Research focuses on high-dimensional statistical analysis, including dimensionality reduction and inference for principal component regression in high dimensions, with interest in connecting statistical structure and inference to real data and decision support (e.g., risk measurement, valuation, and allocation in finance). JST BOOST Next-Generation AI fellow; contract analyst at Nospare Inc.

Education

University of Tsukuba 2025 – 2028 (expected)
Ph.D. in Mathematics, Graduate School of Science, Degree Programs in Pure and Applied Sciences / Aoshima Laboratory

University of Tsukuba 2023 – 2025
M.Sc. in Mathematics / Aoshima Laboratory

Kagoshima University 2019 – 2023
B.Sc. in Mathematics and Informatics, Faculty of Science / Yoshida Laboratory (2022–2023)

Ehime Prefectural Matsuyama Kita High School 2016 – 2019
Graduated (General Course)

Fellowships

Next-Generation AI Human Resource Development Program Fellow Apr. 2025 – Mar. 2028
JST BOOST / University of Tsukuba
Selected through competitive internal screening (12 fellows at Tsukuba). Supported with living-expense-equivalent funding and research funds for theoretical and applied work in high-dimensional statistics.

Work Experience

Analyst (Contract) Jan. 2026 – present
Nospare Inc.
Transitioned from an education-focused internship. Consulting (management, DX, statistics), data analysis and machine learning solutions, and education programs (EDRIGHT, practitioner seminars).

Part-time work Mar. 2019 – Oct. 2024
Food service (front of house & kitchen), liquor retail, logistics & warehousing

Internships

Mitsubishi UFJ Trust Investment Technology Institute (MTEC) Oct. 2026
Internship at an MUFG Group institute developing financial solutions via mathematical and information sciences

Mizuho-DL Financial Technology Nov. 2025
Internship related to management/risk analytics, data analytics R&D, and investment methods

Mizuho Research & Technologies Aug. 2025 (3 days)
AI R&D internship; image recognition workshop on manufacturing inspection

Nospare Inc. Dec. 2024 – Dec. 2025
Education-business internship (EDRIGHT, seminars); later transitioned to contract role

Research Interests

High-dimensional statistics; high-dimension, low-sample-size data; dimensionality reduction (PCA, PLS, CCA, MCA, ICA); machine learning; representation learning; generative models; causal discovery (LiNGAM); text embeddings; finance; open science

Selected Talks

- Sho Sakai, Kazuyoshi Yata, and Makoto Aoshima, “Exploring Principal Component Regression in High-Dimensional Data,” SPRING and BOOST Selected Students Research Presentation, Mar. 2026 (poster).
- Same (extended), Seminars by Alumnae/Alumni of Kagoshima University, Mar. 2026 (oral).
- Same, The 20th Japan Statistical Society Spring Meeting, Mar. 2026 (poster).
- Same (extended), ISACT 2026, Feb. 2026 (poster).
- “Hypothesis testing for PCR coefficients in high-dimensional data,” Mathematical Society of Japan Annual Meeting, Mar. 2025 (oral).

Teaching & Academic Service

Teaching Assistant Oct. 2023 – Mar. 2025
University of Tsukuba
Statistical Exercise & Computer Exercise (Fall 2023, Fall 2024); Computer Mathematics I & Linear Algebra I (Spring 2024)

Vice Chair & Technical Assistant Apr. 2024 – present
User Committee, Department of Mathematics, University of Tsukuba (Chair: Apr. 2025–Mar. 2026)

International Symposium Staff Dec. 2023 / Dec. 2024
Symposia on Large Complex Data

Staff, Mathematics Trial Program 2023 / 2024
University of Tsukuba

Research Assistant Oct. 2023 – Nov. 2024
University laboratory

Qualifications & Courses

- Upper Secondary Teacher’s License (Informatics, Mathematics); Lower Secondary Teacher’s License (Mathematics) (Mar. 2023)
- TOEIC L&R 735 (Mar. 2022)
- Matsuo–Iwasawa Lab (UTokyo) Deep Learning courses: Deep Generative Models (2026 Summer); LLM 1, Fundamentals, Financial Market Trading and ML (2026 Spring)

Awards & Outreach (summary)

- Japanese Red Cross Society Silver Merit Medal (2025); Minister of Health, Labour and Welfare certificate of appreciation for bone marrow donation; Kagoshima University General Education Center Award (2020)
- Bone Marrow Bank Youth Ambassador (2024–); blood donation (2018–)
- Host/organizer, podcast “Data Science LG: Learning Together in Statistics and Data Science”

Links

Website: https://shoshohh.github.io/ note: https://note.com/sho_77 Kaggle: https://www.kaggle.com/sakaisho

Note: Provisional CV compiled from information already published on the personal website. No phone number or postal address is included.